

Studying the Dynamic Behaviour of Chip Components during Reflow Soldering

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Abstract: Since lead-free soldering is initiated in the electronic industry in 1st of July 2006, studying the behaviour of lead-free solders in depth is more important than ever. The small chip components of present day demand for very accurate component placement machines, in order to prevent common reflow failures such as skewing or tombstoning. The ability of components to be self-aligned during soldering works against these failures, therefore it matters to what extent the solder promote this effect. Dynamic behaviour of SMT (Surface Mount Technology) chip components during lead-free reflow soldering is demonstrated in the paper, especially focusing on movement. A force model is given by the virtue of the four forces, which have effect on the chip during reflow soldering, namely: the force originating from hydrostatic and capillary pressure, the force of gravity, the force of dynamic friction and the force originating from the surface tension. The self-alignment of 0603 size chip components has been investigated in our experiment. The experiments were carried out using lasercut stainless steel stencil and SAC305 type lead-free solder paste. The results have shown that the self-alignment of components does occur even in the case of 400-500 μm misplacements.

1. INTRODUCTION

The small chip components of present day (0201 to 01005) demand for very accurate component placement machines, in order to prevent common reflow failures such as skewing or tombstoning. The ability of components to be self-aligned during soldering works against these failures, therefore it matters to what extent the solder promote this effect.

Among the forces, which have effect on the chip components during reflow soldering, only the force originating from the surface tension of the solder has changed significantly with the implementation of lead-free technology; it has grown between the solder and the pad, as well as, between the solder and the atmosphere. Since the force, which is related to surface tension, acts on the component most robustly and forces it to do movement, it can be predicted that component self-alignment does occur with the usage of lead-free solders as well. At the same time several studies and articles have claimed that the self-

alignment capability of components was lower when using lead-free solders.

Since this revelation was in contrast with the observations obtained from our previous experiments, we decided to carry out both theoretical and experimental investigations in this field.

2. THEORETICAL BACKGROUND

First of all the topic of predicting the dynamic behaviour of chip components during reflow soldering should be divided into two parts. The first issue is to determine the shape of the molten solder fillet, while the second issue is to calculate the forces, which are acting on the component, by the known fillet. There are two main theoretical approaches for determining the shape of the molten solder; calculating the equilibrium shape by minimizing the energy of the system or applying the pressure continuity principle.

2.1. Predicting the solder fillet profile by minimizing the energy of the system

The equilibrium shape of a liquid meniscus at a liquid-gas phase boundary of a system - in which solid, liquid, and gaseous phases coexist - is given by a balance of forces acting on the system. In the case when the boundary condition is that the solder wets until the end of the metallization and the contact angle depends on the volume of the solder, the energy of the system which should be minimized is given by equation (1) [1, 2]:

$$E = E_s + E_G, \text{ where:} \quad (1)$$

$$- E_s - \text{energy due to surface tension} = \int_A \gamma_{LG} dA \quad (2)$$

$$- E_G - \text{energy due to gravity} = \int_V \rho \cdot g \cdot z \cdot dxdydz \quad (3)$$

- γ_{LG} – surface tension of the solder-gas boundary
- ρ – solder density
- g – gravitational constant
- A – surface of the fillet
- V – volume of the solder

When the boundary condition is that the end of pad is not reached by the solder, thus the contact angle is equal to the wetting angle, as can be seen in Fig. 1., equation (2) forms as the following [3]:

$$E_s = \int_{A_0} \gamma_{LG} dA + \int_{A_1} -\gamma_{LG} \cdot \cos\theta_1 dA + \int_{A_2} -\gamma_{LG} \cdot \cos\theta_2 dA \quad (4)$$

- A_0 – boundary area of the solder and the gas
- A_1 – boundary area of the solder and the pad
- A_2 – boundary area of the solder and the component metallization
- θ_1 – wetting angle on the contact line of solder and the component metallization
- θ_2 – wetting angle on the contact line of solder and the pad

Actually equation (4) is separated form of equation (2), where the surface of the solder is separated into three parts. The contact line of the solder is moving on the pads toward inside of the solder, decreasing the total energy of the system. Hence the second and the third term have to be subtracted from the first term in equation (4). Since only that component of force (force originating from γ_{LG} surface tension), which is parallel to the contact line movement, doing work on the system, γ_{LG} in second and third term has to be multiplied by $\cos\theta$.

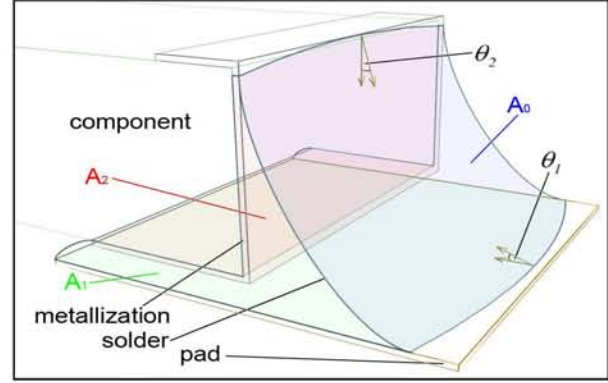


Fig. 1. Boundary surfaces of the molten solder

By solving equation (1) with any of the boundary conditions the shape of the molten solder can be determined. Since the solder joints are quite small, the dominant term in equation (1) is the first one, i.e. the energy due to the surface tension. The forces which are acting on a component can be calculated by the known fillet profile.

2.2. Forces, that are acting on a chip component during reflow soldering

Mainly four forces are acting on a chip component during reflow soldering. The force originating from surface tension is acting on the boundary contact line of solder, gas, and component metallization. The forces originating from hydrostatic and capillary pressure are acting on the area of the metallization of the component; while the force originating from dynamic friction depends on the mass of the liquid solder which should be actuated.

In case of component misplacements (due to the placement machine inaccuracy) the main force, which aids the self-alignment, is originating from the surface tension of the liquid solder. In a three-phase system, where gaseous-solid-liquid phases coexists (Fig. 2.); the balance between the surface tensions can be, described by the Young equation (5):

$$\gamma_{SG} = \gamma_{LS} + \gamma_{LG} \cdot \cos\theta \quad (5)$$

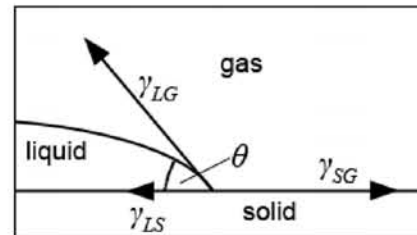


Fig. 2. Surface tensions of a three-phase system

The terms of the Young equation are the followings:

- γ_{SG} – surface tension of the gas-solid boundary
- γ_{LS} – surface tension of the liquid-solid boundary
- γ_{LG} – surface tension of the gas-liquid boundary
- θ – contact angle

The sum of these forces, which are originating from the surface tension, acts on the appointment place of the solder, gas, and component metallization. In general case the appointment place of a three-phase system is a space curve, which is called as contact line in soldering technologies. Therefore the resultant force originating from the surface tension can be obtained by integrating the terms due to surface tension along the contact line (6):

$$\vec{F}_{st} = \sum_i^{LS, LG, SG} \int \gamma_i d\vec{l} \quad (6)$$

Basically the degree of component misplacement can be described by three values:

- offset in direction parallel to component shorter side (lateral, x direction)
- offset in direction parallel to component longer side (y direction)
- rotational, theta offset (θ offset)

The forces, originating from hydrostatic- and capillary pressures of the molten solder, push the component out from the solder. These forces are acting on the vertical side- and bottom side metallization of the component as illustrated by Fig. 3. (F_h and F_c). In symmetrical cases (no misplacement, only lateral misplacement) the forces, which are acting on the vertical side metallization, are opposite to each other, thus their effect is compensated, but in the case of y direction misplacements the forces due to hydrostatic- and capillary pressures aid the y direction alignment.

The capillary pressure is the pressure difference between the two sides of a curved liquid surface. The pressure drop across the fillet (ΔP), can be determined by the Laplace's equation (eq. 7). In its most general form the equation is [4]:

$$\Delta P = \gamma \cdot \left(\frac{1}{r_1} + \frac{1}{r_2} \right), \quad (7)$$

where r_1 and r_2 are the radii of curvature of the fillet, measured normal to the surface in mutually

orthogonal planes. The hydrostatic pressure inside a liquid can be described by eq. (8) in general form:

$$P_h = \rho_l \cdot g \cdot m, \text{ where} \quad (8)$$

- ρ_l – density of the liquid
- g – is the gravitational constant
- m – the height of the liquid column

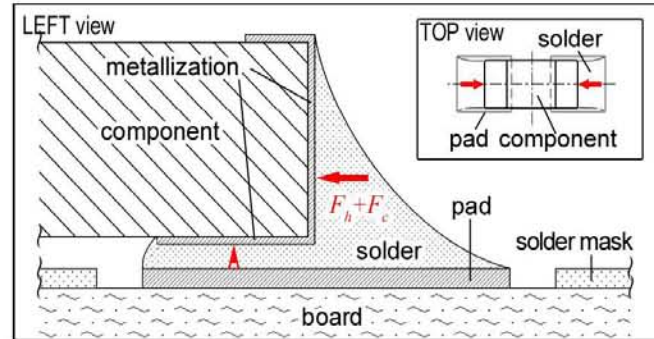


Fig. 3. Forces due to hydrostatic- and capillary pressures

The principle of the pressure continuity is that in a static solder fillet, no pressure gradients exist horizontally and the pressure in the vertical direction varies proportionally to the distance from the fillet surface, thus it seems that the m height of the liquid column is balanced by the ΔP capillary pressure. [4]

Therefore the capillary pressure can be calculated by any point, which takes place above than the s line ($n_i \in A_i$), as illustrated by Fig. 4. The s line is infinitesimally close to the bottom side metallization of the component. The force originating from hydrostatic- and capillary pressures (F_p) can be determined by integrating the pressure values along the surface of the component metallization (9):

$$\vec{F}_p = \int_{A_{cs}} \left(\rho_s \vec{g} \cdot m(\vec{r}) + \gamma_{LG} \left(\frac{1}{r_1(b(\vec{r}))} + \frac{1}{r_2(b(\vec{r}))} \right) \right) \cdot d\vec{A} \quad (9)$$

- A_{cs} – surface of component metallization
- ρ_s – density of the molten solder
- $m(\vec{r})$ – height of the liquid column, which is infinitesimally close to the point designated by the $\vec{r}$ vector on the A_{cs} surface
- $b(\vec{r})$ – point on the top of the liquid column, where the capillary pressure should be calculated

The dynamic friction between the liquid and the solid phases slows the movement of the liquid phase. Hence the dynamic friction of a liquid is actually the viscosity of that liquid. In our case the dynamic friction reduce the self-alignment of the component.

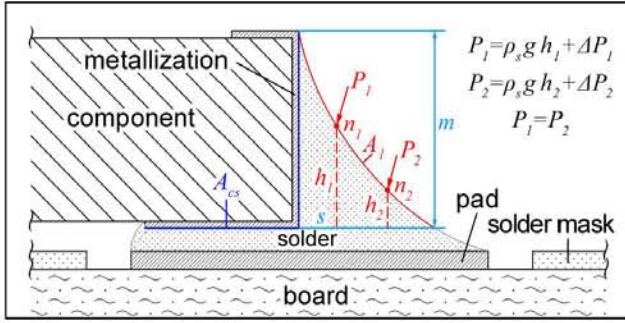


Fig. 4. Principle of pressure continuity

The liquid (molten solder) can be imagined as a series of horizontal layers. The top layer in the molten solder is infinitesimally close to the bottom side metallization of the component, and its speed is equal to the speed of the component movement. The speed of bottom layer is 0, equal to the speed of the pad. Anywhere inside the molten solder, the speed of an upper layer is higher than the speed of a lower layer; microscopically the molecules of the upper layer are colliding with the molecules of the lower layer and a part of the faster molecules kinetic energy transforms to heat energy. The force, originating from the dynamic friction between a solid and a liquid phase, can be described by the Newton equation (10) [5]:

$$F = \eta \cdot A \frac{dv}{dz} \quad (10)$$

In our case, when the component is moving in the molten solder, the solid phase is the metallization of the component; therefore the surface of the metallization should be taken into consideration. Thus the force (11), which is decelerating the movement of the component, can be written as the following:

$$\vec{F}_v = \int_{A_{cs}} \frac{\eta_s \cdot (\vec{v} - \vec{v}_0)}{d(\vec{r})} dA, \text{ where:} \quad (11)$$

- A_{cs} – surface of the component metallization
- η_s – viscosity of the solder
- $\vec{v}$ – speed vector of the component movement
- $\vec{v}_0$ – speed of the point at d distance, which is 0 if it is on the pad, and not 0 if the point is on the surface of the molten solder
- $d(\vec{r})$ – distance between the pad and the point under investigation, which is designated by the $\vec{r}$ vector on the A_{cs} surface

Consequently the resultant force (12), which acts on the component during reflow soldering, is the sum of the above described forces and the gravity force:

$$\vec{F}_{sum} = \vec{F}_{st} + \vec{F}_p - \vec{F}_v + \vec{F}_{grav} \quad (12)$$

$$\begin{aligned} \vec{F}_{sum} = & \sum_i^{LS, LG, SG} \int_v \gamma_i d\vec{l} + \\ & + \int_{A_{cs}} \left(\rho_s \vec{g} \cdot m(\vec{r}) + \gamma_{LG} \left(\frac{1}{r_1(b(\vec{r}))} + \frac{1}{r_2(b(\vec{r}))} \right) - \frac{\eta_s \cdot (\vec{v} - \vec{v}_0)}{d(\vec{r})} \right) dA + \\ & + \int_{V_{comp}} \rho_{comp} \vec{g} \cdot dV \end{aligned}$$

3. BACKGROUND OF SIMULATION

For giving adequate prediction of the aligning force, a 3D profile calculation is necessary. A method based on the principle of energy minimum was developed. This method based on equation (1) and on the fact whereby the energy of the fillet should be minimized. In our previous work [6] we presented a mesh based method. That algorithm had two unique properties compared to other methods. The first is, that algorithm starts with coarse mesh and refines the mesh during its running. The second one is that the base of profile calculation is two dimensional but the algorithm is able to solve the 3D problem by applying calculations in alternating direction. That algorithm was suitable for our previous examination, but the starting with coarse mesh is not applicable, when studying finer misplacements of chip components is necessary.

The original algorithm can not handle fine mesh, because the number of the degree of freedoms will be large during the minimalization of the energy function of the surface profile. Thus, the surface profiles have to be described by a method, which describes the curve with less parameter. This problem exists long time ago in the field of the technical computing, so there are numerous solutions. For example, applying Spline functions is possible to describe the solder profile [7]. Among the possibilities the Bèzier-curve was chosen to describe the profile. The essence of the Bèzier-curves is that a complicated curve can be described by the aid of less parameter. This method is very useful in design or in computer graphic. The definition of the x and y coordinates in the case of cubic Bèzier-curve is the following (13) [8]:

$$\begin{aligned} x(t) &= a_x t^3 + b_x t^2 + c_x t + x_0, \text{ where} \\ y(t) &= a_y t^3 + b_y t^2 + c_y t + y_0 \end{aligned} \quad (13)$$

- $t \in [0, 1]$ is the normalized arc length.

The parameters can be obtained by equations (14-15) [8]:

$$\begin{aligned} a_x &= x_3 - x_0 - c_x - b_x & a_y &= y_3 - y_0 - c_y - b_y \\ b_x &= 3(x_2 - x_1) - c_x & \text{and } b_y &= 3(y_2 - y_1) - c_y \\ c_x &= 3(x_1 - x_0) & c_y &= 3(y_1 - y_0) \end{aligned} \quad (14-15)$$

where the x and y coordinates are the coordinates of the end points ($P_0(x_0, y_0)$, $P_3(x_3, y_3)$) and the focal points ($P_1(x_1, y_1)$, $P_2(x_2, y_2)$), as can be seen in Fig. 5.

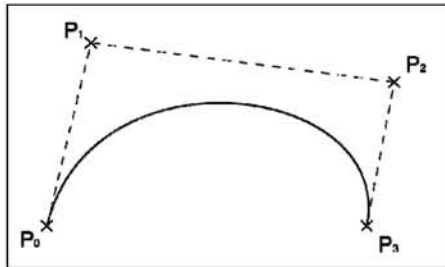


Fig. 5. Control and focal points of the Bézier-curve

As can be seen, the number of parameters, which describes the curve, is irrespective from the applied mesh density. Consequently, the number of degrees of freedom is irrespective from the applied mesh density as well. Moreover, the direction of the line determined by the end- and the focal points can be used for determining the contact angle of the solder. This advantage does not exist when Spline functions are applied.

Naturally it is necessary to examine, whether the Bézier-curve is applicable for the liquid profile calculations, and what kind of degree is necessary for suitable fitting. For this examination a cross-section image of a chip resistor was applied (Fig. 6.). With the aid of the Matlab Image processing toolbox (Roberts method was applied as edge detection algorithm, and Loess function was applied as smoothing method on the obtained edges) the solder profile was converted into vectors, thus the necessary calculations were realizable.

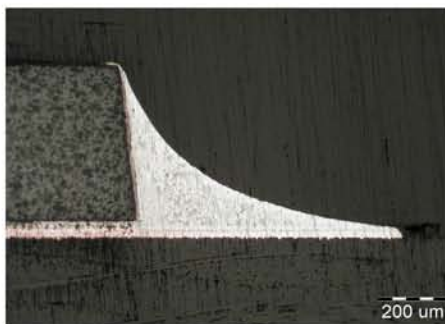


Fig. 6. Half section of a chip component

The result of the examination is illustrated by Fig. 7. It can be seen, that the quadratic Bézier-curve fits to the solder profile very well, thus it is suitable for describing the solder profile, it is not necessary to apply higher order. In future we would like to apply Bézier-curves as fillet profiles, and after determining them, the forces, which are acting on the component during reflow soldering, can be calculated.

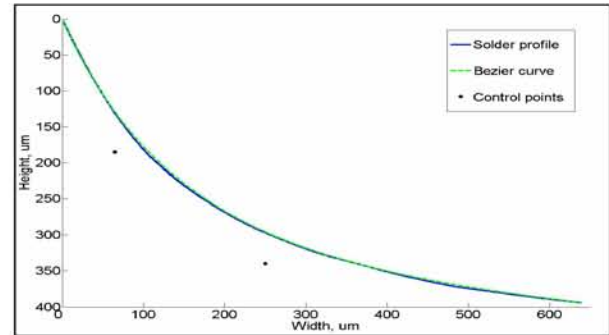


Fig. 7. Accommodation of the Bézier Curve to the profile

4. EXPERIMENT CONCERNING COMPONENT SELF-ALIGNMENT

In our experiment the self-alignment of 0603 size (1,5x0,75x0,4 mm) chip components was investigated. The components were misplaced programmed in x and y directions, and theta offset was applied as well. The placement offsets in x direction were 0 to 900 μm in 100 μm steps, while the y placement offsets were 0, 280 and 380 μm . Besides the x - y offsets, the theta offsets were 0°, 10° and 20°.

Senju M705-GRN360-K-V (type 4, SAC305) lead-free solder paste and 150 μm thick, laser-cut, stainless steel stencil was used for the experiment, while the applied reflow profile is illustrated by Fig. 8.

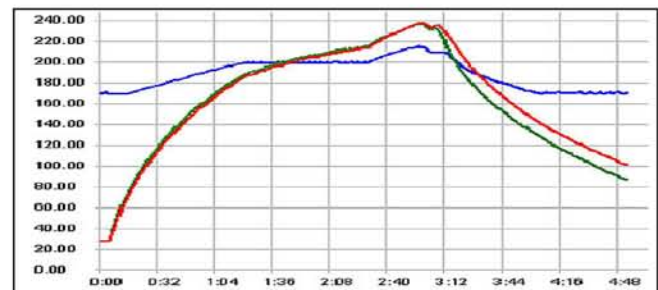


Fig. 8. Reflow profile of the experiment

The exact place of components before- and after reflow soldering can be measured by the guidelines of the IPC-9850 standard, which recommends placing fiducial points around the component (Fig. 9.).

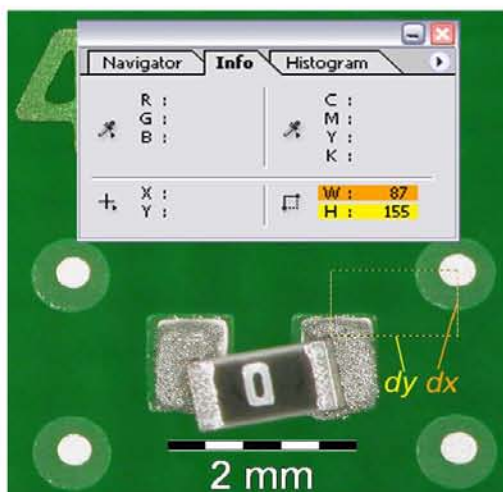


Fig. 9. Measuring the place of the component

By evaluating the results it can be said, that the self-alignment of chip components has occurred even in the case of 500 μm x direction misplacement. When large y direction- and theta offsets were applied, the self-alignment of components has occurred only in the case of lower x direction misplacement, as expected. Fig. 10 shows the x misplacement ranges (up to 500 μm) where self-alignment occurred with given y and theta offset parameters, while Table 1. gives information about the distances, which are traveled by the chip components during reflow soldering.

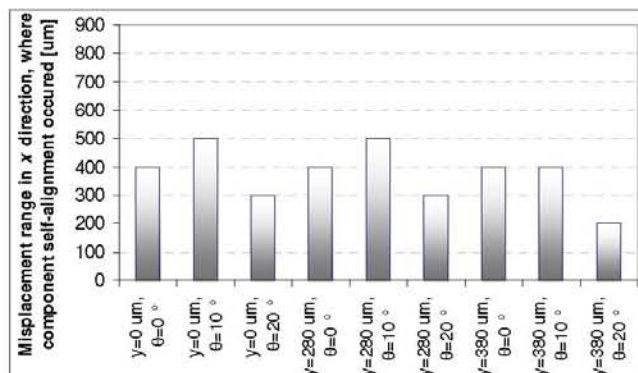


Fig. 10. Occurrence of component self-alignment in the case of different x , y and θ offsets

	Place of comp. before reflow soldering			Distance covered during reflow by the comp.		
	x_{offset} [μm]	y_{offset} [μm]	θ_{offset} [°]	Δx [μm]	Δy [μm]	$\Delta \theta$ [°]
R121	74	-74	-0,2	48	0	0,3
R136	571	-22	12,0	338	52	11,0
R173	281	69	21,9	177	57	15,7

Tab. 1. Distance traveled by the component during soldering

5. SUMMARY

The self-alignment of 0603 size chip components has been investigated in our experiment. Pick&Place machine was programmed to place the components with 0...900 μm x and y offsets and 0...20° rotational (θ) offsets. According to the results it can be said that the self-alignment of chip components has occurred in the case of lateral (x direction) misplacements up to 500 μm . The results have also revealed that the chip components move in y direction much less than in x direction, because the forces, which are originating from hydrostatic and capillary pressures (acting on the vertical metallization of the component and moving it in y direction), are opposite to each other and the chip component is driven by their difference only.

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